

RCM — Reasoning and Correction Module

OOF™ Origin Open Foundation™

Independent Methodology Authority

Category: AI & Interpretation

Module: Reasoning and Correction Module

Type: Integrity & Validation Module

Parent Standard: CLIA® — Cognitive Layer Integrity Standard

Version: 1.0

Status: Canonical · Open Module

Effective Date: 21 April 2026

Authority: OOF™ Origin Open Foundation™

Protection: MIP™ — Methodological Intellectual Property

Canonical Language: English (UCL™)

Canonical Definition

Reasoning and Correction defines the non-bypassable conditions under which a system's reasoning remains logically consistent, and errors are detected and corrected during operation.

A system is valid only if reasoning does not produce contradictions and incorrect reasoning is identified and corrected.

A. Module Abstract

This module establishes the conditions required for stable and reliable reasoning.

It ensures that:

- reasoning remains consistent
- contradictions are prevented
- errors are detected
- correction is enforced

B. Core Principle

Reasoning must remain consistent and self-correcting.

C. Core Conditions

A system is valid only if all of the following conditions are satisfied:

- Reasoning remains logically consistent and free of contradiction.
 - Contradictions are detected and cannot persist.
 - Errors in reasoning are identified during operation.
 - Incorrect reasoning is corrected before influencing outcomes.
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D. Non-Bypassable Rule

All reasoning and correction conditions are non-bypassable.

No reasoning process may operate without consistency and correction.

E. Validity State

A system is valid only if all reasoning remains consistent and all detected errors are corrected.

F. Invalid State

A system is invalid if:

- reasoning produces contradictions
 - contradictions are not resolved
 - errors are not detected
 - incorrect reasoning influences outcomes
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G. Scope

This module applies to:

- AI reasoning systems
 - decision-making systems
 - autonomous agents
 - multi-agent reasoning environments
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H. Methodology

Reasoning integrity is enforced during operation.

Correction must occur as part of the reasoning process, not after it.

I. MIF — Minimum Implementation Framework

A system implementing Reasoning and Correction must:

- enforce logical consistency in reasoning
 - detect contradictions during processing
 - identify errors in reasoning paths
 - apply correction before output generation
 - maintain traceability of reasoning steps
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J. Use Cases

Use Case 1 — Contradictory Decision Logic

Without this module:

A system produces outputs based on conflicting reasoning paths.

With this module:

Contradictions are detected and corrected before output.

Result:

Reasoning remains consistent and reliable.

Use Case 2 — Adaptive Learning System

Without this module:

The system repeats incorrect reasoning patterns over time.

With this module:

Errors are detected and corrected during operation.

Result:

The system improves without reinforcing incorrect logic.

Canonical Closing Statement

If reasoning is not consistent, it is not valid.

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